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Automatically assigning natural language labels to non-conforming behavior of processes

Abstract

Systems and methods for automatically assigning labels to one or more types of non-conforming behavior of execution of a process are provided. An aligned process defining non-conforming behavior of execution of a process is received. One or more types of the non-conforming behavior of the execution of the process is identified from the aligned process. Labels identifying the one or more types are assigned to the non-conforming behavior. The labels assigned to the non-conforming behavior are output.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates generally to process mining, and more particularly to automatically assigning natural language labels to non-conforming behavior in processes.
BACKGROUND
(2) Processes are sequences of activities executed by one or more computers to perform various tasks. In process mining, conformance checking is performed to evaluate whether the actual execution of the process conforms to the expected execution of the process. Conventionally, conformance checking is performed by manually comparing an event log representing the actual execution of the process with a process model representing the expected execution of the process. However, such conventional conformance checking is a time-consuming and labor-intensive process.
BRIEF SUMMARY OF THE INVENTION
(3) In accordance with one or more embodiments, systems and methods for automatically assigning labels to one or more types of non-conforming behavior of execution of a process are provided. An aligned process defining non-conforming behavior of execution of a process is received. One or more types of the non-conforming behavior of the execution of the process is identified from the aligned process. Labels identifying the one or more types are assigned to the non-conforming behavior. The labels assigned to the non-conforming behavior are output. In one embodiment, the

process is an RPA (robotic process automation) process.
(4) In one embodiment, the labels are generated according to a standardized format to identify the one or more types of non-conforming behavior. The labels assigned to the one or more types of the non-conforming behavior may be displayed with the aligned process.
(5) In one embodiment, a non-conforming skipped activity is identified in the aligned process where the activity has an outgoing log-only path and an outgoing model-only path. The outgoing log-only path occurs in an event log of the process as outgoing from the activity but does not occur in a process model of the process and the outgoing model-only path occurs in a process model of the process as outgoing from the activity but does not occur in the event log. In another embodiment, a non-conforming repeated activity in the aligned process is identified where the activity has a model-only edge that is both outgoing and incoming. The model-only edge occurs in a process model of the process but does not occur in an event log of the process. In another embodiment, a non-conforming loop back to an earlier point in the aligned process is identified where a node of the aligned process has an outgoing log-only edge to a previously traversed node. The outgoing log-only edge occurs in an event log of the process but does not occur in a process model of the process.
(6) In one embodiment, the non-conforming behavior is identified as being a block comprising a sub-process of the aligned process. The labels for the block may be generated according to a standardized format to identify the one or more types of the non-conforming behavior based on a name of an activity in the block.
(7) These and other advantages of the invention will be apparent to those of ordinary skill in the art by reference to the following detailed description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows an illustrative process on which conformance checking may be performed in accordance with one or more embodiments;
(2) FIG. **2** shows a method for automatically assigning natural language labels to one or more types of non-conforming behavior of a process, in accordance with one or more embodiments;
(3) FIG. **3** shows an exemplary process model, in accordance with one or more embodiments;
(4) FIG. **4** shows an exemplary event log of a process, in accordance with one or more embodiments;
(5) FIG. **5** shows an exemplary aligned process, in accordance with one or more embodiments;
(6) FIG. **6** shows a portion of an aligned process showing non-conforming skipped activity behavior, in accordance with one or more embodiments;
(7) FIG. **7** shows a portion of an aligned process showing non-conforming repeated activities behavior, in accordance with one or more embodiments;
(8) FIG. **8** shows a portion of an aligned process showing non-conforming loop back behavior, in accordance with one or more embodiments;
(9) FIG. **9** shows an exemplary user interface presenting labels assigned to types of non-conforming behavior, in accordance with one or more embodiments; and
(10) FIG. **10** is a block diagram of a computing system according to an embodiment of the invention.
DETAILED DESCRIPTION
(11) FIG. **1** shows an illustrative process **100** on which conformance checking may be performed, in accordance with one or more embodiments. Process **100** may be applied to perform any suitable task, such as, e.g., document processing. In one embodiment, process **100** may be implemented as an RPA (robotic process automation) process for automatically performing a task using one or more

RPA robots. However, it should be understood that process **100** may be any suitable process that can be modelled as a workflow, such as, e.g., a business workflow.

(12) Process **100** comprises activities **102-116**, which represent predefined steps in process **100**. As shown in FIG. **1**, process **100** is modeled as a directed graph where each activity **102-116** is represented as a node and each transition between activities **102-116** is represented as edges linking the nodes. The transition between activities represents the execution of process **100** from a source activity to a destination activity. Process **100** starts at start activity **102** and proceeds to activity A **104**. Process **100** then proceeds, in parallel, to a first branch comprising activity B **106** and activity C **108** and a second branch comprising activity D **110** and activity E **112**. Process **100** then proceeds to activity F **114** and ends at end activity **116**. Execution of process **100** is recorded as an event log. Each event of the event log refers to the execution of an activity for a certain case at a certain point in time. The event may be represented as a tuple comprising an activity, a case identifier, and a time stamp. Expected execution of process **100** is modelled as a process model.

(13) In process mining, conformance checking is performed on, e.g., process **100** to evaluate whether actual execution of process **100** (as identified in the event log) conforms to the expected execution of process **100** (as identified in the process model). In accordance with embodiments of the present invention, types of non-conforming behavior of the execution of process **100** are automatically assigned natural language labels. Advantageously, such automatic assignment of natural language labels to the types of non-conforming behavior facilitates user understanding and analysis of the non-conforming behavior.

(14) FIG. **2** shows a method **200** for automatically assigning natural language labels to one or more types of non-conforming behavior of a process, in accordance with one or more embodiments. Method **200** may be performed by one or more suitable computing devices, such as computer **900** of FIG. **9**.

(15) At step **202** of FIG. **2**, an aligned process defining non-conforming behavior of execution of a process is received. As used herein, non-conforming behavior refers to actual behavior of a process that does not conform to the expected behavior of the process. The aligned process may have been generated based on an event log representing actual execution of the process and a process model representing expected execution of the process by aligning the event log to the process model. Generation of an aligned process is further described in U.S. Patent Application Publication No. 2021/0200574, entitled “Visual Conformance Checking of Processes,” filed Dec. 30, 2019, the disclosure of which is incorporated herein by reference in its entirety. An exemplary process model is shown in FIG. **3**, an exemplary event log is shown in FIG. **4**, and exemplary aligned process is shown in FIG. **5**, which are described in detail below.

(16) The aligned process may be received by loading the aligned process from a storage or memory of a computer system or by receiving the aligned process from a remote computer system. In one example, the process is process **100** of FIG. **1**. In one embodiment, the process is an RPA process automatically executed by one or more RPA robots.

(17) FIG. **3** shows an exemplary process model **300**, in accordance with one or more embodiments. Process model **300** represents expected execution of process **100** of FIG. **1**. Process model **300** is modelled as directed graph where each activity of the process is represented as a node and the execution of the process from a source activity to a destination activity is represented as an edge connecting the nodes representing the source activity and the destination activity. Each edge in process model **300** is associated with a number representing a frequency of execution of that edge.

(18) Process model **300** models expected execution of process **100** using gateways to represent diversions in process **100**. The gateways control how the process flows during execution. Gateways are represented in process model **300** as gateway nodes. For example, gateway nodes, shown in FIG. **3** as diamond-shaped nodes identified with a “+”, may represent parallel relationships. Process model **300** may also comprise gateway nodes representing other types of relationships, such as, e.g., exclusive choice relationships, looping relationships, sequential relationships, etc. As

shown in FIG. **3**, parallel split gateway **302** represents splitting of the path from activity A **104** into a first path to activity B **106** and a second path to activity D **110** to concurrently execute activity B **106** and activity D **110**, and parallel join gateway **304** represents the joining of a first path from activity C **108** and a concurrent second path from activity E **112** into a single path to activity F **114**.
(19) FIG. **4** shows an exemplary event log **400** of a process, in accordance with one or more embodiments. Event log **400** records execution of process **100** of FIG. **1** and will be described with reference to FIG. **1**. Event log **400** is formatted as a table having rows **402** and columns **404**. As shown in FIG. **4**, rows **402** of event log **400** comprises rows **402**-A through **402**-F, each corresponding to an event defining execution of a respective activity **104-114**, with a particular time stamp and with a particular case identifier. Event log **400** may also include rows **402** for start activity **102** and end activity **116** in some embodiments. Columns **404** of event log **400** comprises column **404**-A identifying the activity, column **404**-B identifying the time stamp, and column **404**-C identifying the case identifier. Columns **404** may include additional columns identifying additional attributes of activities.
(20) FIG. **5** shows an exemplary aligned process **500** of a process, in accordance with one or more embodiments. Aligned process **500** comprises aligned events and transitions that represent the actual execution of the process aligned over the process model of the process. Aligned process **500** is modelled as directed graph where each activity of the process is represented as a node and the execution of the process from a source activity to a destination activity is represented as an edge connecting the nodes representing the source activity and the destination activity. Each edge in aligned process **500** is associated with a number representing a frequency of execution of that edge.
(21) When the actual execution of certain paths of the process matches the expected execution of the certain paths of the process, execution of the certain paths of the process is identified in aligned process **500** as being conforming behavior. When the actual execution of certain paths of the process deviates from the expected execution of the certain paths of the process, execution of the certain paths of the process is identified in aligned process **500** as being non-conforming behavior. The non-conforming behavior may be non-conforming log only behavior where paths occur in the event log but not in the process model or the non-conforming behavior may be non-conforming model-only behavior where paths occur in the process model but not in the event log. In one embodiment, aligned process **500** identifies the behavior by color coding the nodes and/or edges of aligned process **500** to identify conforming behavior (e.g., as blue), the non-conforming log-only behavior (e.g., as orange), and the non-conforming model-only behavior (e.g., as green).
(22) At step **204** of FIG. **2**, one or more types of the non-conforming behavior of the execution of the process are identified from the aligned process. The non-conforming behavior may comprise, for example, non-conforming paths comprising activities, edges, or a sub-process of the process. In one embodiment, the non-conforming behavior may comprise one or more of skipped activities, repeated activities, and loop backs. Other types of non-conforming behavior are also contemplated, such as, e.g., out of order performance of activities, violation of exclusive relationships, performing (at least parts of) multiple branches in an exclusive block, etc.
(23) In one embodiment, the non-conforming behavior is identified in the aligned process based on rule-based pattern matching. The rule-based pattern matching identifies patterns of log-only and model-only edges.
(24) In one embodiment, a non-conforming skipped activity is identified where a particular activity has an outgoing log-only path and an outgoing model-only path The outgoing log-only path is a path (comprising, e.g., one or more edges and/or nodes) that occurs in the event log as outgoing from the particular activity but does not occur in the process model. The outgoing model-only path is a path that occurs in the process model as outgoing from the particular activity but does not occur in the event log. When the particular activity has an outgoing log-only path and an outgoing model-only path, it is assumed that the particular activity is skipped. FIG. **6** shows a portion of an aligned process **600** showing a non-conforming skipped activity, in accordance with one or more

embodiments. In aligned process **600**, the “approve invoice” activity **604** is skipped during the instances of execution of edges **602**.
(25) In one embodiment, non-conforming repeated activity behavior is identified where a particular activity has a log-only edge that is both outgoing and incoming, thereby forming a self-loop. The log-only edge is an edge that occurs in the event log but does not occur in the process model. FIG. **7** shows a portion of an aligned process **700** showing non-conforming repeated activities behavior, in accordance with one or more embodiments. In aligned process **700**, the “approve invoice” activity **704** is repeated during instances of execution of edges **702**.
(26) In one embodiment, non-conforming loop back behavior is identified where a particular node (e.g., a gateway node or an activity node) has an outgoing log-only edge to a node previously traversed during that instance of execution. The outgoing log-only edge is an outgoing edge that occurs in the event log but does not occur in the process model. FIG. **8** shows a portion of an aligned process **800** showing non-conforming loop back behavior, in accordance with one or more embodiments. In aligned process **800**, execution of the process loops back to an earlier point in the aligned process **800** and activities are repeated during instances of execution of edges **802**.
(27) It should be understood that the identification of non-conforming behavior is not limited to the rule-based pattern matching discussed above. For example, in one embodiment, non-conforming behavior can be identified using pattern recognition techniques for graphs to identify a set of predefined patterns associated with a specific type of non-conforming behavior. In another embodiment, a machine learning based model may be trained to identify non-conforming behavior using non-conformance behavior that has been previously identified. In a further embodiment, the non-conforming behavior can be identified while performing the process alignment to generate the aligned process. The particular approach for identifying non-conforming behavior may be determined based on the type of non-conforming behavior being detected.
(28) At step **206** of FIG. **2**, labels identifying the one or more types are assigned to the non-conforming behavior. The labels are natural language labels that allow users to easily identify and explore non-conforming behavior of interest. The labels may be generated using any suitable approach.
(29) In one embodiment, the labels are generated using a standardized format based on the type of the non-conforming behavior to provide natural language labels. For example, where the type of the non-conforming behavior is identified as being a non-conforming skipped activity, the label is generated as: “skipped <name>”, where <name> refers to the name of the activity that is skipped. Where the type of the non-conforming behavior is identified as being a non-conforming repeated activity, the label is generated as: “repeating <name>”, where <name> refers to the name of the activity that is repeated. Where the type of the non-conforming behavior is identified as being a non-conforming loop back, the label is generated as: “loop back from <name1> to <name2>”, where <name1> refers to the name of the from-activity and <name2> refers to the name of the to-activity. If the type of the non-conforming behavior cannot be determined (e.g., where the rules for identifying the types of the non-conforming behavior do not cover all possible non-conforming behavior or where the machine learning model cannot recognize all non-conforming behavior), a label of “unknown non-conforming behavior” label is assigned to the non-conforming behavior. As such, labels are assigned to all non-conforming behavior. Only fully conforming behavior have no labels.
(30) In many cases, the non-conforming behavior is not limited to a single activity or node but may comprise a block comprising a sub-process of the process. To assign labels to such blocks, the block hierarchy of the process tree of the process is utilized, which may be derived from the process model during process alignment. A standardized format based on a type of behavior in the block is provided to generate a label of the block as follows: “<type> block containing <name>”, where <type> is a type of behavior in the block (e.g., parallel) and <name> is a name of any activity in the block. For some processes (or process apps), it is desirable to assign process-specific

names to the blocks, which can be done at the app-level. In some embodiments, blocks may be manually labeled by a user.

(31) At step **208** of FIG. **2**, the labels assigned to the non-conforming behavior are output. In one embodiment, the labels are output by displaying the labels to a user on a display device, thereby facilitating selection and exploration of non-conforming behavior of interest by a user. In other embodiments, the labels may be output by, for example, storing the labels on a memory or storage of a computer system or by transmitting the labels to a remote computer system.

(32) FIG. **9** shows an exemplary user interface **900** presenting labels assigned to non-conforming behavior, in accordance with one or more embodiments. The assigned labels may have been assigned according to method **200** of FIG. **2**. As shown in FIG. **9**, an aligned process **904** is shown with a table **902** listing labels identifying types of non-conforming behavior of aligned process **904** in a first row and a corresponding number of cases in a second row. In user interface **900**, a user can select (e.g., by clicking) on the labels, which will select all cases that are assigned to the selected label. The user can then further explore this selection and perform root-cause analysis to find out what may cause these deviations to occur. Advantageously, the labels identifying a type of non-conforming behavior enable the user to select and explore non-conforming behavior of interest of process **902**.

(33) FIG. **10** is a block diagram illustrating a computing system **1000** configured to execute the methods, workflows, and processes described herein, including FIG. **2**, according to an embodiment of the present invention. In some embodiments, computing system **1000** may be one or more of the computing systems depicted and/or described herein. Computing system **1000** includes a bus **1002** or other communication mechanism for communicating information, and processor(s) **1004** coupled to bus **1002** for processing information. Processor(s) **1004** may be any type of general or specific purpose processor, including a Central Processing Unit (CPU), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Graphics Processing Unit (GPU), multiple instances thereof, and/or any combination thereof. Processor(s) **1004** may also have multiple processing cores, and at least some of the cores may be configured to perform specific functions. Multi-parallel processing may be used in some embodiments.

(34) Computing system **1000** further includes a memory **1006** for storing information and instructions to be executed by processor(s) **1004**. Memory **1006** can be comprised of any combination of Random Access Memory (RAM), Read Only Memory (ROM), flash memory, cache, static storage such as a magnetic or optical disk, or any other types of non-transitory computer-readable media or combinations thereof. Non-transitory computer-readable media may be any available media that can be accessed by processor(s) **1004** and may include volatile media, non-volatile media, or both. The media may also be removable, non-removable, or both.

(35) Additionally, computing system **1000** includes a communication device **1008**, such as a transceiver, to provide access to a communications network via a wireless and/or wired connection according to any currently existing or future-implemented communications standard and/or protocol.

(36) Processor(s) **1004** are further coupled via bus **1002** to a display **1010** that is suitable for displaying information to a user. Display **1010** may also be configured as a touch display and/or any suitable haptic I/O (input/output) device.

(37) A keyboard **1012** and a cursor control device **1014**, such as a computer mouse, a touchpad, etc., are further coupled to bus **1002** to enable a user to interface with computing system. However, in certain embodiments, a physical keyboard and mouse may not be present, and the user may interact with the device solely through display **1010** and/or a touchpad (not shown). Any type and combination of input devices may be used as a matter of design choice. In certain embodiments, no physical input device and/or display is present. For instance, the user may interact with computing system **1000** remotely via another computing system in communication therewith, or computing

system **1000** may operate autonomously.

(38) Memory **1006** stores software modules that provide functionality when executed by processor(s) **1004**. The modules include an operating system **1016** for computing system **1000** and one or more additional functional modules **1018** configured to perform all or part of the processes described herein or derivatives thereof.

(39) One skilled in the art will appreciate that a “system” could be embodied as a server, an embedded computing system, a personal computer, a console, a personal digital assistant (PDA), a cell phone, a tablet computing device, a quantum computing system, or any other suitable computing device, or combination of devices without deviating from the scope of the invention. Presenting the above-described functions as being performed by a “system” is not intended to limit the scope of the present invention in any way, but is intended to provide one example of the many embodiments of the present invention. Indeed, methods, systems, and apparatuses disclosed herein may be implemented in localized and distributed forms consistent with computing technology, including cloud computing systems.

(40) It should be noted that some of the system features described in this specification have been presented as modules, in order to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very large scale integration (VLSI) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as field programmable gate arrays, programmable array logic, programmable logic devices, graphics processing units, or the like. A module may also be at least partially implemented in software for execution by various types of processors. An identified unit of executable code may, for instance, include one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located together, but may include disparate instructions stored in different locations that, when joined logically together, comprise the module and achieve the stated purpose for the module. Further, modules may be stored on a computer-readable medium, which may be, for instance, a hard disk drive, flash device, RAM, tape, and/or any other such non-transitory computer-readable medium used to store data without deviating from the scope of the invention. Indeed, a module of executable code could be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules, and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set, or may be distributed over different locations including over different storage devices, and may exist, at least partially, merely as electronic signals on a system or network.

(41) The foregoing merely illustrates the principles of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the disclosure and are included within its spirit and scope. Furthermore, all examples and conditional language recited herein are principally intended to be only for pedagogical purposes to aid the reader in understanding the principles of the disclosure and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Moreover, all statements herein reciting principles, aspects, and embodiments of the disclosure, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future.

Claims

1. A computer-implemented method comprising: executing, by one or more computing devices, an RPA (robotic process automation) process by one or more RPA robots to generate an event log of the execution; receiving, by the one or more computing devices, an aligned process comprising events and transitions, representing actual execution of the RPA process as identified in the event log, that are aligned over a process model representing expected execution of the RPA process, the aligned process defining behavior of the actual execution that is non-conforming to the process model; identifying, by the one or more computing devices, one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process; assigning, by the one or more computing devices, natural language labels identifying the one or more types to the non-conforming behavior using a standardized format based on 1) the one or more types of the non-conforming behavior and 2) names of one or more activities of the RPA process that are non-conforming; displaying, by the one or more computing devices, the aligned process with the natural language labels assigned to the non-conforming behavior on a display device; receiving, by the one or more computing devices, user input of a selection of one of the natural language labels via the display device; and in response to the user input, automatically selecting, by the one or more computing devices, cases having the non-conforming behavior assigned to the selected natural language label.
2. The computer-implemented method of claim 1, wherein identifying, by the one or more computing devices, one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying a non-conforming skipped activity in the aligned process where the activity has an outgoing log-only path and an outgoing model-only path, wherein the outgoing log-only path occurs in an event log of the RPA process as outgoing from the activity but does not occur in a process model of the RPA process, and wherein the outgoing model-only path occurs in the process model as outgoing from the activity but does not occur in the event log.
3. The computer-implemented method of claim 1, wherein identifying, by the one or more computing devices, one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying a non-conforming repeated activity in the aligned process where the activity has a model-only edge that is both outgoing and incoming, wherein the model-only edge occurs in a process model of the RPA process but does not occur in an event log of the RPA process.
4. The computer-implemented method of claim 1, wherein identifying, by the one or more computing devices, one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying a non-conforming loop back to an earlier point in the aligned process where a node of the aligned process has an outgoing log-only edge to a previously traversed node, wherein the outgoing log-only edge occurs in an event log of the RPA process but does not occur in a process model of the RPA process.
5. The computer-implemented method of claim 1, wherein identifying, by the one or more computing devices, one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying the non-conforming behavior as being a block comprising a sub-process of the aligned process.
6. The computer-implemented method of claim 5, wherein assigning, by the one or more computing devices, natural language labels identifying the one or more types to the non-conforming behavior comprises: generating the natural language labels for the block according to the standardized format to identify the one or more types of the non-conforming behavior based on names of the one or more activities in the block.
7. An apparatus comprising: a memory storing computer instructions; and at least one processor configured to execute the computer instructions, the computer instructions configured to cause the at least one processor to perform operations of: executing an RPA (robotic process automation)

process by one or more RPA robots to generate an event log of the execution; receiving an aligned process comprising events and transitions, representing actual execution of the RPA process as identified in the event log, that are aligned over a process model representing expected execution of the RPA process, the aligned process defining behavior of the actual execution that is non-conforming to the process model; identifying one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process; assigning natural language labels identifying the one or more types to the non-conforming behavior using a standardized format based on 1) the one or more types of the non-conforming behavior and 2) names of one or more activities of the RPA process that are non-conforming; displaying the aligned process with the natural language labels assigned to the non-conforming behavior on a display device; receiving user input of a selection of one of the natural language labels via the display device; and in response to the user input, automatically selecting cases having the non-conforming behavior assigned to the selected natural language label.
8. The apparatus of claim 7, wherein identifying one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying a non-conforming skipped activity in the aligned process where the activity has an outgoing log-only path and an outgoing model-only path, wherein the outgoing log-only path occurs in an event log of the RPA process as outgoing from the activity but does not occur in a process model of the RPA process, and wherein the outgoing model-only path occurs in the process model as outgoing from the activity but does not occur in the event log.
9. The apparatus of claim 7, wherein identifying one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying a non-conforming repeated activity in the aligned process where the activity has a model-only edge that is both outgoing and incoming, wherein the model-only edge occurs in a process model of the RPA process but does not occur in an event log of the RPA process.
10. The apparatus of claim 7, wherein identifying one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying a non-conforming loop back to an earlier point in the aligned process where a node of the aligned process has an outgoing log-only edge to a previously traversed node, wherein the outgoing log-only edge occurs in an event log of the RPA process but does not occur in a process model of the RPA process.
11. A non-transitory computer-readable medium storing computer program instructions, the computer program instructions, when executed on at least one processor, cause the at least one processor to perform operations comprising: executing an RPA (robotic process automation) process by one or more RPA robots to generate an event log of the execution; receiving an aligned process comprising events and transitions, representing actual execution of the RPA process as identified in the event log, that are aligned over a process model representing expected execution of the RPA process, the aligned process defining behavior of the actual execution that is non-conforming to the process model; identifying one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process; assigning natural language labels identifying the one or more types to the non-conforming behavior using a standardized format based on 1) the one or more types of the non-conforming behavior and 2) names of one or more activities of the RPA process that are non-conforming; displaying the aligned process with the natural language labels assigned to the non-conforming behavior on a display device; receiving user input of a selection of one of the natural language labels via the display device; and in response to the user input, automatically selecting cases having the non-conforming behavior assigned to the selected natural language label.
12. The non-transitory computer-readable medium of claim 11, wherein identifying one or more types of the non-conforming behavior of the actual execution of the RPA process from the aligned process comprises: identifying the non-conforming behavior as being a block comprising a sub-

process of the aligned process.
13. The non-transitory computer-readable medium of claim 12, wherein assigning natural language labels identifying the one or more types to the non-conforming behavior comprises: generating the natural language labels for the block according to the standardized format to identify the one or more types of the non-conforming behavior based on names of the one or more activities in the block.